Unraveling Telangana's Path to Inclusive Development

A Proposal report for the BDM capstone Project

Submitted by

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<u>Declaration Statement</u>

I am working on a Project titled "Unraveling Telangana's Path to Inclusive Development". I extend my appreciation to Open Data Unit, Government Of Telangana for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate :

Name : Jishnu S G

Date : 13-05-2025

1. Executive Summary

The state of Telengana leads in data-driven governance, harmonizing the industrial growth and agrarian resilience through initiatives like Open Data Telangana, TS-iPASS and Rythu Bandhu. Transparencies in the policies empower stakeholders to leverage datasets for equitable and sustainable development.

Despite these advancements, challenges persist in ensuring inclusive growth. The delayed approvals for small-scale enterprises might discourage local entrepreneurship and thereby limit employment rates and economic activities. The digitalization gap in the land revenue system (if existing) indicates underutilization of the digital infrastructure and hinder governance efficiency. On top of it, the erratic rainfall pattern might drive mismatched agricultural investment in mechanization could inflate the farm costs. These issues risk constraining economic inclusivity, climate resilience and might widen urban-rural division.

The study employs time–series analysis of industrial approval timelines, correlation between rainfall with agri-vehicle registrations, and comparative study for the e-Stamp adoption to find the trends and patterns. Python tools like Pandas and Plotly will be used for data processing of different datasets for generating district-level insights.

The expected outcomes will help to prioritize districts for fast-tracked business approvals, not just only for small-scaled ones even for the large ones, climate-aligned agricultural supports and infrastructure for farmers and scaling the digitized governance to ensure states growth balances industrial efficiency with agrarian sustainability.

2. Organization Background

Telangana is a state in the south-central part of the Indian subcontinent adorned with rich history, diverse culture and rapid growth. It is the eleventh largest and twelfth most populated state in India as per the 2011 census.

The state is notable for its progressive approach to data accessibility and transparency through its initiative Open Data Telangana. With an Open Data Policy in place, the government ensures that data related to various domains is made easily accessible to the public. This also aligns with the Digital India mission and Digital Telangana, where the aim is to empower the people and stakeholders digitally literate.

Open Data Telangana is testament to the state's commitment towards the growth of the state by leveraging the digital infrastructure. It empowers citizens, researchers and business stakeholders by providing valuable information, thereby encouraging social and economic development with citizen-centric governance by ensuring the ethos of transparency.

3. Problem Statement

1. Small investments face systemic approval delays : Smaller projects may experience disproportionately longer approval timelines compared to large-scale proposals, discouraging local entrepreneurship.
2. e-Stamp adoption efficiency over document registration : Identify if digital transitions in registrations improved revenue collection efficiency for scalable governance and promoting digital inclusion.
3. Rainfall-mechanization mismatch inflates farm costs : Uneven rainfall forces over/under investments in agri-vehicles without promising irrigation infrastructure might lead to unnecessary capital expenditure and subsidies.

Supporting Articles and NEWS Readings

4. Background of the Problem

Telangana is a semi-arid climate area with hot and dry conditions. These conditions positioned Telangana's strong agricultural outputs and positioned it as a leading producer of rice, maize, and grains. But from the recent past the state is well known for the foreign and domestic investments, in the fiscal year 2021-22 alone the state attracted Rs 17,867 core worth investments.

The state initiated various projects like TS-iPASS (Telangana State Industrial Project Approval and Self-Certification System), T-Hub (Technology Hub), Rythu Bandhu, Telangana Electric Vehicles and Energy Storage Policy from 2015 onwards to help both the industrial and agricultural parties. In November, 2015 at the inauguration ceremony of T-Hub Sri K T Rama Rao mentioned "our government slogan has been innovate, incubate and incorporate", and years downline investments in the state is increasing day by day.

This project will explore the various aspects of the growth starting from the road transport sales, TS-iPASS and land registrations across the state to interpret the revenue growth and the influence of weather in socio-economic activity of state's industrial and agricultural sectors.

According to the media houses, the state's major investments and rapid growth are on the Hyderabad side, being known for agricultural robustness, we will be trying to figure out the Urban-Rural Divide (if any) and the impact of rapid development on climate change (if any) from the available data.

5. Problem Solving Approach

This project uses data-driven methodologies to analyze systemic approval delays for large and small enterprises, digitalization gap in the land revenue and mismatched agricultural mechanization using Telengana's district-level data from 2019 January to 2024 December. The study will utilize time series analysis and comparative analytics to derive actionable insights.

Methods

To evaluate approval delays for enterprises, time-series analysis tracks the duration between application submission and approval across industries and demographics. Statistical methods like mean, median mode will help to identify the patterns linking delays to variables such as investment size, demographics and social status of the applicants. For digitalization efficiency, trend analysis compares manual and digital revenue streams over the time, assessing the adoption rate and the processing efficiency. To address the climate-mechanization mismatch finding, monthly correlation with rainfall patterns with agricultural registrations will be utilized. This can help us in classifying districts with profiles based on climate risks and mechanization levels. Over these, cross-cutting techniques like data integration, to merge industrial, weather and transport datasets to capture cross-sector insights such as seasonal demand spikes in agricultural machinery and change in trend in vehicle ownership can be revealed.

Data

The analysis relies on four key datasets:

1. Industrial approvals: Daily records of investment applications, approvals, and applicant demographics, aggregated into monthly district-level summaries.
2. Weather metrics: Monthly rainfall, temperature, and humidity data for all districts.
3. Land registrations: Revenue and transaction counts from manual and digital processes.
4. Vehicle sales: Agricultural and non-agricultural vehicle registrations categorized by type, fuel, and capacity.

Analysis Tools

- Python with Pandas : For preliminary data processing and cleaning and analysis python might be required due to the volume of data, and Pandas can handle data more comprehensively for data cleaning.
- Python with Matplotlib/Seaborn : It will be useful to create visual representations, also, might use the Plotly to have interactive visual plots.

Since we have multiple data related to multiple sectors, we have to analyze them separately. Also we have the district name or location name as common actress the data, combining them might fetch much more insights over uni-sector analysis.

Dataset Link : Google Drive | Open Data Telangana Portal

6. Expected Timeline

6.1 Work Breakdown Structure :

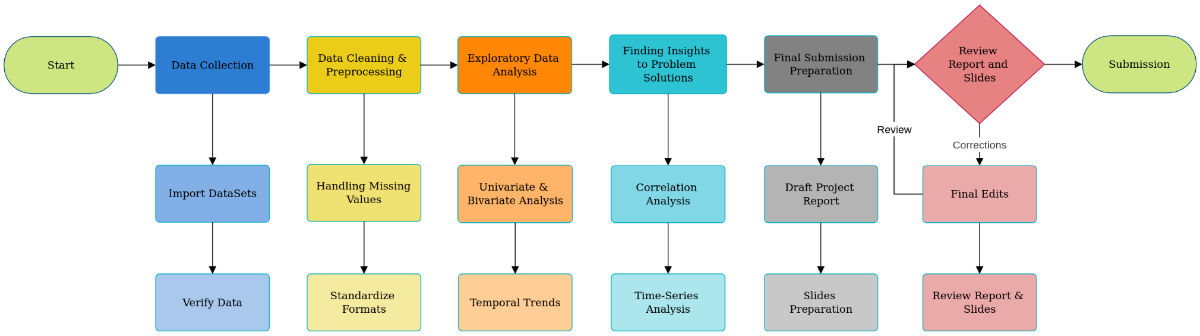

Figure 1 Work BreakDown Structure for completion of project.

6.2 Gantt chart :

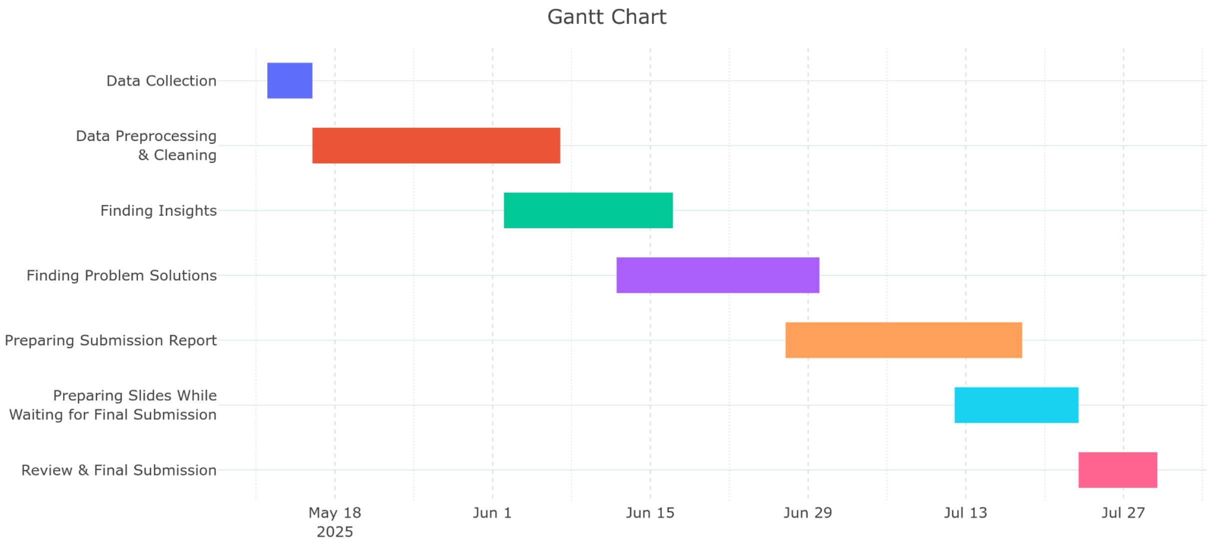

Figure 2 Expected timeline for completion of project.

7. Expected Outcome

- A better idea for food processing and infrastructure industries to be aligned with specific districts, based on climate and economic conditions for better growth.
- Help the governments to allocate subsidies or infrastructure to support farmers and their crops using schemes like Rythu Bandhu,for having higher agricultural productivity.
- Identifying the districts that need better vehicle infrastructure, more EV charging stations and logistic hubs to keep up with the economic development.
- To assist the government to prioritize districts needing road updates or public transport expansion based on the vehicle sales trends.
- Evidence based policy recommendations to fast-track approval for small investments to reduce the delays and boost equitable industrial developments.
- Identify the sectors and business activities that are facing longest approval delays, investigate links to investment size and district specificity.